

Section 2 Information, Martingales, and Multi-Period Markets

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Overview

Section 1 priced claims in a market that trades once. Real markets trade again and again, and information arrives between trades. This section builds the three objects a repeated market needs. It then upgrades the fundamental theorem.

- §2.1: a model of *information over time*, called a *filtration*.
- §2.2: *conditional expectation*, the forecast made with that information.
- §2.3: *martingales*, the processes that are fair games with respect to that information.
- §2.4: the fundamental theorem in many periods. A market is arbitrage-free exactly when discounted prices are martingales under some equivalent measure.

A finite event tree is used to picture information and conditional expectations. The section then states the multi-period fundamental theorem, which is the bridge to continuous-time pricing. Proofs are worked in class in the gray blocks and appear in the complete version of these notes. The binomial model returns in Section 5, where it becomes a numerical method.

2.1 Information and Filtrations

This subsection builds the model of “what is known when.” The economic constraint is simple. A trading decision made at date t may use only information available at date t .

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. In the finite models of this section Ω is the set of all complete paths the economy can follow from date 0 to date T . A single $\omega \in \Omega$ is therefore not a local outcome at one date. It is one entire history. The σ -algebra $\mathcal{F}$ collects the events that can be distinguished once everything has happened, and $\mathbb{P}$ assigns their physical probabilities. Write $\Omega = \{\omega_1, \dots, \omega_M\}$, and write $\pi_\omega = \mathbb{P}(\{\omega\}) > 0$ for the probability of a state. Every state is given strictly positive probability.

At an intermediate date the path is only partly revealed. We know which group of states is still possible. We do not know which single state will occur. The language for that is a σ -algebra, so we define one first.

Definition 2.1 (Events and σ -algebras).

A collection $\mathcal{G}$ of subsets of Ω is a σ -algebra on Ω if

- (i) $\Omega \in \mathcal{G}$;
- (ii) $A \in \mathcal{G} \implies A^c \in \mathcal{G}$; and
- (iii) $A_1, A_2, \dots \in \mathcal{G} \implies \bigcup_{n \geq 1} A_n \in \mathcal{G}$.

The members of $\mathcal{G}$ are called events. For any collection C of subsets of Ω , $\sigma(C)$ denotes the smallest σ -algebra containing C .

Condition (iii) is the countable clause, and it is what the σ in the name refers to. On a finite Ω it costs nothing: closure under pairwise unions already gives it, because there are only finitely many subsets to begin with. Section 3 works on an infinite state space, and there the countable version is the one that matters.

Read a σ -algebra as a list of questions. If $A \in \mathcal{G}$, then the question “did A happen?” is one that the holder of $\mathcal{G}$ can answer. Condition (ii) says that answering that question also answers “did A fail to happen?”. Condition (iii) says that two answerable questions can be combined into one.

Example 2.2. Let $\Omega = \{1, 2, 3, 4\}$.

- (a) Decide whether each of these is a σ -algebra on Ω :

$$\mathcal{G}_1 = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}, \quad \mathcal{G}_2 = \{\emptyset, \{1\}, \{1, 2\}, \Omega\}.$$

- (b) Write out $\sigma(C)$ for $C = \{\{1\}, \{2\}\}$.

Definition 2.3 (Filtration).

A filtration is a family $\mathbb{F} = (\mathcal{F}_t)_{t=0}^T$ of sub- σ -algebras of $\mathcal{F}$ with

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \text{whenever } s \leq t.$$

The σ -algebra $\mathcal{F}_t$ represents the information available at date t : if $A \in \mathcal{F}_t$, then by date t an observer can determine whether A has occurred.

Information grows over time. A question that could be answered yesterday can still be answered today. That is the entire content of the nesting condition, and it is the reason the family is called increasing.

For the models used in this section we assume two further things. The first is that $\mathcal{F}_0 = \{\emptyset, \Omega\}$, so nothing is known at date 0. The second is that $\mathcal{F}_T = \mathcal{F}$, so everything is known at the end. These are convenient assumptions for the event-tree model. They are not part of the general definition of a filtration.

On a finite state space there is a picture that goes with $\mathcal{F}_t$. The σ -algebra groups together the terminal paths that date- t information cannot yet tell apart. Those groups have a definition of their own. A *cell*, or *atom*, of $\mathcal{F}_t$ is a non-empty event in $\mathcal{F}_t$ that contains no smaller non-empty event of $\mathcal{F}_t$. The cells partition Ω . Every event in $\mathcal{F}_t$ is a union of cells, and knowing which cell holds the true state is exactly what date- t information tells you. As information arrives the cells become finer. Listing the cells is the practical way to write a filtration down, and the next example does it.

Example 2.4. A die is rolled once, so $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the outcome is reported in stages. At date 1 you learn whether the number is odd or even. At date 2 you learn in addition whether it is at most 3 or at least 4. At date 3 you learn the number itself. Write down the information cells at each date, check that they refine, and count the events in each $\mathcal{F}_t$.

Work through the dates. At date 0 nothing has been reported, so there is one cell. At date 1 the report splits Ω by parity. At date 2 each of those is split again by the second report. At date 3 everything is known.

$$\begin{aligned} t = 0 : & \quad \{1, 2, 3, 4, 5, 6\} \\ t = 1 : & \quad \{1, 3, 5\}, \quad \{2, 4, 6\} \\ t = 2 : & \quad \{1, 3\}, \quad \{5\}, \quad \{2\}, \quad \{4, 6\} \\ t = 3 : & \quad \{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\} \end{aligned}$$

They refine, because $\{1, 3, 5\} = \{1, 3\} \cup \{5\}$ and $\{2, 4, 6\} = \{2\} \cup \{4, 6\}$, and every date-2 cell is a union of singletons. So $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3$, and this is a filtration.

Each $\mathcal{F}_t$ is every union of its cells, so a date with k cells has 2^k events: 2, 4, 16 and 64.

Two features of this example are worth naming, and both are why a three-date report teaches more than a two-date one. The cells at date 2 are not all the same size. And on two of them, $\{5\}$ and $\{2\}$, the outcome is already fully known at date 2, a date earlier than the last one. Information does not have to arrive at the same rate along every path.

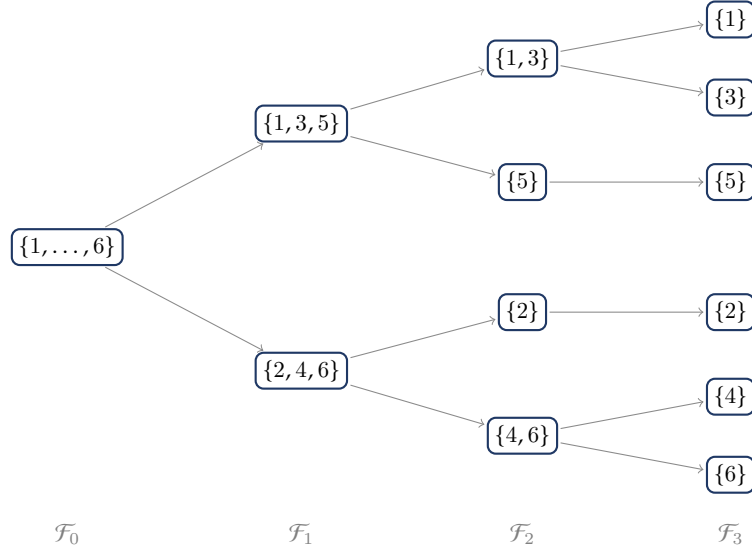


Figure 1: The die filtration of Example 2.4. Each box is one information cell, and each column is one date. Reading left to right, a cell either splits or stays whole. The cells $\{5\}$ and $\{2\}$ do not split again, because on those outcomes the number is already known at date 2.

Definition 2.5 (Measurable, adapted).

Let $\mathcal{F}_t$ be a sub- σ -algebra of $\mathcal{F}$.

(i) A random variable Z is $\mathcal{F}_t$ -measurable if

$$\{Z \in B\} \in \mathcal{F}_t \quad \text{for every interval } B \subseteq \mathbb{R}.$$

(ii) A process $X = (X_t)_{t=0}^T$ is adapted to $\mathbb{F}$ if X_t is $\mathcal{F}_t$ -measurable for every t .

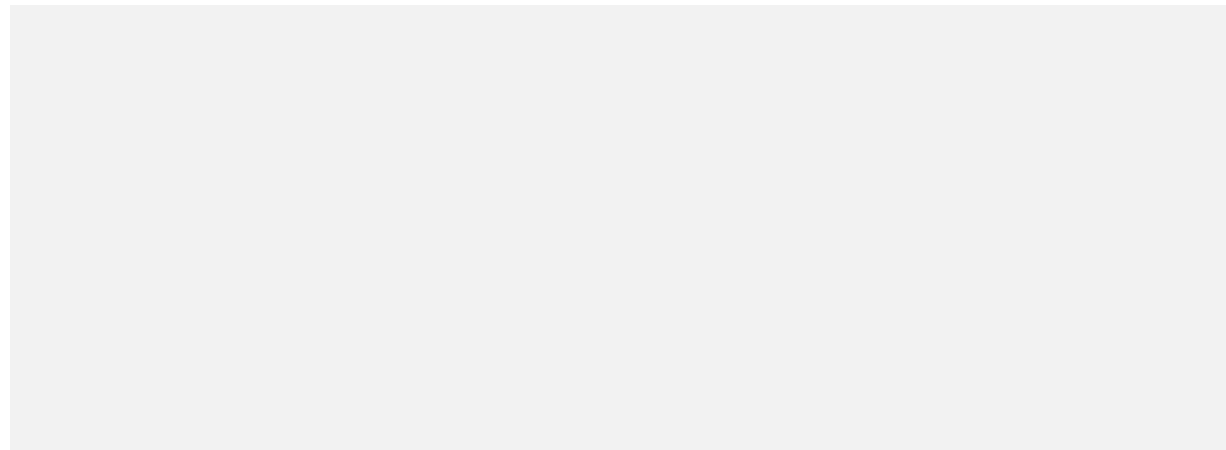
Condition (i) is one idea written once: every question about the value of Z can be answered from date- t information. On a finite Ω it says exactly that Z is constant on every date- t cell, and that is the form used in every calculation below. The general definition asks the same of every Borel set rather than every interval. The two agree, because the intervals generate the Borel sets, and nothing in this course turns on the difference.

The finite-state form is worth checking once. Suppose Z is constant on every cell. Then $\{Z = c\}$ is the union of the cells carrying the value c , so it is an event in $\mathcal{F}_t$. Suppose instead that every set $\{Z = c\}$ lies in $\mathcal{F}_t$. These sets partition Ω and each is a union of cells, so every cell sits inside exactly one of them. Then Z is constant on every cell.

Keep the two words apart. Measurability is a property of one random variable at one date. Adaptedness is a property of a whole process against a whole filtration. So a random variable is $\mathcal{F}_t$ -measurable, and a process is adapted to $\mathbb{F}$.

Adaptedness is the mathematical form of “no clairvoyance.” Asset prices are adapted, because the market quotes them in real time. A trading strategy must respect the same rule. Let θ_t be the portfolio chosen at date t , after $\mathcal{F}_t$ has been observed, and held to date $t + 1$. Then θ_t must be $\mathcal{F}_t$ -measurable. Section 2.4 fixes that indexing convention and uses it. Most definitions in this course are bookkeeping that enforces this one rule.

Example 2.6. Stay with the die filtration of Example 2.4. Let X be the number rolled. Decide which of the following are $\mathcal{F}_2$ -measurable, and say why: (a) X itself; (b) $Y = \mathbf{1}\{X \text{ is even}\}$; (c) W , the smallest number still possible at date 2. Then name one event that lies in $\mathcal{F}_2$ but not in $\mathcal{F}_1$.



Our running example is the two-period binomial tree. At each date the market moves up (u) or down (d). So $\Omega = \{uu, ud, du, dd\}$. Figure 2 shows the filtration. At $t = 0$ all four paths are possible. At $t = 1$ the first move is known. At $t = 2$ the whole path is known.

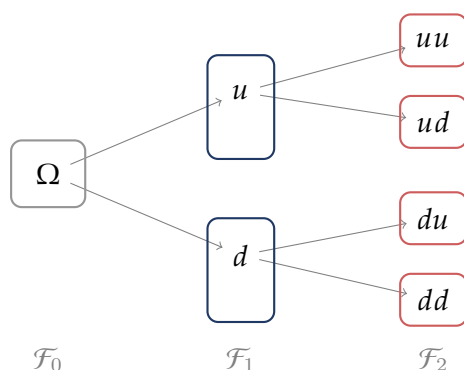


Figure 2: The filtration of the two-period tree. A box is one information cell. At $t = 0$ the observer can answer nothing beyond “did something happen?”, so $\mathcal{F}_0$ has the single cell Ω . At $t = 1$ the observer can tell the two paths starting with u from the two starting with d , but cannot yet separate uu from ud . At $t = 2$ every path is separated, so $\mathcal{F}_2$ has four cells. A process adapted to this filtration takes one value per cell at each date.

2.2 Conditional Expectation

This subsection defines the forecast operator. It gives the average of a future quantity, computed with the information available now. It is the most used tool in the course.

The definition is stated through two properties rather than a formula. That choice is deliberate. The formula that works on a finite tree breaks on the path space of Section 3, and the two properties do not.

Definition 2.7 (Conditional expectation).

Let X be an integrable random variable and let $\mathcal{F}_t$ be a sub- σ -algebra of $\mathcal{F}$. A random variable Y is a version of the conditional expectation $\mathbb{E}[X \mid \mathcal{F}_t]$ if

- (i) Y is $\mathcal{F}_t$ -measurable; and
- (ii) $\mathbb{E}[\mathbf{1}_A Y] = \mathbb{E}[\mathbf{1}_A X]$ for every event $A \in \mathcal{F}_t$,

where $\mathbf{1}_A$ is the indicator of A , worth 1 on A and 0 off it. Such a Y exists, and any two versions agree with probability one. For the finite models of this section both facts are checked just below. They hold for any integrable X , but the general proof needs the Radon–Nikodym theorem, which this course does not build.

Read the two conditions economically. The first says that the forecast may use only information available at date t . The second says that on every event already distinguishable at date t , the forecast preserves the average of X . Those two demands are enough to pin the forecast down completely.

Note what kind of object this is. The forecast $\mathbb{E}[X \mid \mathcal{F}_t]$ is not a number. It is a random variable, and it is $\mathcal{F}_t$ -measurable. Its value depends on which date- t cell the true state falls in. This generalizes the conditional expectation of Statistics I. There we conditioned on one event. Here we condition on a whole information structure.

In the finite models of this section the forecast is computed cell by cell. Write $\pi(C) = \sum_{\omega \in C} \pi_\omega$ for the probability of a cell. On each date- t cell C the forecast takes the constant value

$$\mathbb{E}[X \mid C] = \frac{\sum_{\omega \in C} \pi_\omega X(\omega)}{\pi(C)},$$

the probability-weighted average of X over the states still possible. Every calculation in this section uses that formula.

It is worth seeing in one line why this formula is the object the definition describes. It is constant on each cell, so it is $\mathcal{F}_t$ -measurable. And any event $A \in \mathcal{F}_t$ is a disjoint union of cells, so summing the cell averages over those cells and weighting by $\pi(C)$ recovers $\mathbb{E}[\mathbf{1}_A X]$ exactly. Both conditions hold, so this is the conditional expectation. Nothing else is: on a single cell C condition (ii) forces the constant value to be the ratio above, and that fixes it everywhere.

The formula is a computation rule for a finite model. The definition is the pair of conditions, and that pair is what survives into continuous time.

Example 2.8. Return to the die filtration of Example 2.4 and let X be the number rolled. Compute $\mathbb{E}[X \mid \mathcal{F}_1]$ and $\mathbb{E}[X \mid \mathcal{F}_2]$. Then check the tower property between the two, and check that both average to $\mathbb{E}[X]$.

Apply the formula cell by cell. Every state has probability $\frac{1}{6}$, so inside a cell the weights are equal and each forecast is a plain average.

At date 1 the cells are $\{1, 3, 5\}$ and $\{2, 4, 6\}$:

$$\mathbb{E}[X \mid \{1, 3, 5\}] = \frac{1 + 3 + 5}{3} = 3, \quad \mathbb{E}[X \mid \{2, 4, 6\}] = \frac{2 + 4 + 6}{3} = 4.$$

At date 2 the cells are $\{1, 3\}$, $\{5\}$, $\{2\}$ and $\{4, 6\}$:

$$\mathbb{E}[X \mid \{1, 3\}] = 2, \quad \mathbb{E}[X \mid \{5\}] = 5, \quad \mathbb{E}[X \mid \{2\}] = 2, \quad \mathbb{E}[X \mid \{4, 6\}] = 5.$$

Compare the two forecasts. The date-1 forecast takes two values, 3 and 4. The date-2 forecast takes values 2 and 5. More information gives a forecast that moves further from the overall mean, because it separates outcomes the coarser one had to average together.

Now the tower property, one date-1 cell at a time. Inside $\{1, 3, 5\}$ the date-2 cells are $\{1, 3\}$ with probability $\frac{2}{6}$ and $\{5\}$ with probability $\frac{1}{6}$, so their weights within the cell are $\frac{2}{3}$ and $\frac{1}{3}$:

$$\frac{2}{3}(2) + \frac{1}{3}(5) = \frac{4}{3} + \frac{5}{3} = 3 = \mathbb{E}[X \mid \{1, 3, 5\}].$$

Inside $\{2, 4, 6\}$ the weights are $\frac{1}{3}$ on $\{2\}$ and $\frac{2}{3}$ on $\{4, 6\}$, giving $\frac{1}{3}(2) + \frac{2}{3}(5) = 4$, which again matches.

Finally average each forecast over its own cells. Both date-1 cells have probability $\frac{1}{2}$, so $\frac{1}{2}(3) + \frac{1}{2}(4) = 3.5$. The date-2 cells have probabilities $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$, giving $\frac{1}{3}(2) + \frac{1}{6}(5) + \frac{1}{6}(2) + \frac{1}{3}(5) = 3.5$. Both land on $\mathbb{E}[X] = 3.5$, the mean of a fair die. Proposition 2.9 below says this is no accident.

Proposition 2.9 (Properties of conditional expectation).

For integrable random variables X, Y , dates $s \leq t$, and a bounded $\mathcal{F}_t$ -measurable Z (so that ZX is integrable too):

(a) Linearity:

$$\mathbb{E}[aX + bY \mid \mathcal{F}_t] = a \mathbb{E}[X \mid \mathcal{F}_t] + b \mathbb{E}[Y \mid \mathcal{F}_t].$$

(b) Taking out what is known:

$$\mathbb{E}[ZX \mid \mathcal{F}_t] = Z \mathbb{E}[X \mid \mathcal{F}_t].$$

(c) Tower property (law of iterated expectations):

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}[X \mid \mathcal{F}_s], \text{ and in particular } \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t]] = \mathbb{E}[X].$$

Forecasts respect sums and multiples. A factor already known at date t can be moved outside the forecast. A forecast of a later forecast is just the earlier forecast. You cannot improve today's estimate by averaging estimates you have not seen yet.

Proof.

□

It is worth saying where this is going, before more dice. Soon X will be the future value of a derivative. Its value today will be a forecast of that future value, taken under a reweighted measure written Q :

$$V_t = \frac{1}{1+r} \mathbb{E}^Q[V_{t+1} \mid \mathcal{F}_t].$$

Nothing so far justifies this equation. Here r is the one-period riskless rate, and Q is not defined until §2.4 (Definition 2.17). It is written down here so that the forecast operator has a visible purpose. Everything between now and there is built to make this equation true.

Example 2.10. Four states, $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$, with probabilities $\pi = (0.4, 0.2, 0.3, 0.1)$. The date-1 information has two cells, $A = \{\omega_1, \omega_2\}$ and $B = \{\omega_3, \omega_4\}$. Let $X = (10, 4, 8, 2)$ and $Z = (3, 3, 5, 5)$, both listed state by state. (a) Compute $\mathbb{E}[X \mid \mathcal{F}_1]$ and check the tower property. (b) Check that $\mathbb{E}[ZX \mid \mathcal{F}_1] = Z \mathbb{E}[X \mid \mathcal{F}_1]$. (c) Repeat part (b) with $Z' = (3, 5, 3, 5)$ and say what goes wrong.

Those numbers also show defining condition (ii) at work, and it is worth seeing it once. Write $Y = \mathbb{E}[X \mid \mathcal{F}_1]$, worth 8 on A and 6.5 on B . On the event A ,

$$\mathbb{E}[\mathbf{1}_A X] = 0.4(10) + 0.2(4) = 4.8, \quad \mathbb{E}[\mathbf{1}_A Y] = 0.6(8) = 4.8.$$

On the event B , $\mathbb{E}[\mathbf{1}_B X] = 0.3(8) + 0.1(2) = 2.6$ and $\mathbb{E}[\mathbf{1}_B Y] = 0.4(6.5) = 2.6$. The forecast and the variable have the same average over each event that date-1 information can already decide. That is condition (ii), on numbers.

It is worth saying now why the definition was written the way it was. Section 3 replaces the finite state space by a set of continuous paths, and $\mathcal{F}_t$ records the path seen so far. The cell holding the true state is then every path agreeing with what has been seen, and that set has probability zero. Both sums in the cell formula above are then zero, so the cell formula reads 0/0 and computes nothing. The two conditions of Definition 2.7 never divide, so they still work. The finite formula is a convenience; the characterization is the definition.

The same reasoning applies to measurability. On path space the sets matching an exact history again have probability zero, so “constant on every cell” is not usable. The Borel condition in Definition 2.5 is, and it says the same thing: Z is $\mathcal{F}_t$ -measurable when every question about Z can be answered from date- t information. The rules in Proposition 2.9 continue to hold there. We do not re-prove them. Their proofs need integration theory this course does not assume.

Example 2.11. On the tree of Figure 2 let $S_0 = 100$, $u = 1.2$, $d = 0.8$ and $r = 0$, so $S_2 \in \{144, 96, 96, 64\}$, and let the physical probability of an up move be $p = 0.6$ at each date, independently. Compute $\mathbb{E}[S_2 | \mathcal{F}_1]$ and verify the tower property $\mathbb{E}[\mathbb{E}[S_2 | \mathcal{F}_1]] = \mathbb{E}[S_2]$.

2.3 Martingales in Discrete Time

This subsection names the processes whose forecasts never move. Such a process is the mathematical form of a fair game. After a change of measure it is also the mathematical form of an arbitrage-free price.

Definition 2.12 (Martingale).

An adapted, integrable process $X = (X_t)_{t=0}^T$ is a martingale with respect to the filtration $\mathbb{F}$ and the measure $\mathbb{P}$ if for every t ,

$$\mathbb{E}[X_{t+1} | \mathcal{F}_t] = X_t.$$

If instead $\leq$ holds it is a supermartingale; if $\geq$, a submartingale.

Given everything known today, tomorrow's expected value equals today's value. The game is fair. Note that the property is defined against two things at once, the filtration and the measure. Change either one and a martingale can stop being one. That flexibility is what the rest of the course trades on. A supermartingale drifts down, so it is an unfavorable game. A submartingale drifts up.

Example 2.13. A gambler bets \$1 on each toss of a coin. The coin comes up heads with probability p , independently across tosses. Let X_t be the gambler's cumulative winnings after t tosses, with $X_0 = 0$, and let $\mathcal{F}_t$ be the information in the first t tosses. Decide what kind of process X is for $p = \frac{1}{2}$, for $p = 0.6$ and for $p = 0.4$.

Each toss adds +1 or -1 to the total. The toss at date $t + 1$ is independent of everything known at date t . So

$$\mathbb{E}[X_{t+1} | \mathcal{F}_t] = X_t + p(+1) + (1 - p)(-1) = X_t + (2p - 1).$$

For $p = \frac{1}{2}$ the added term is 0. So X is a martingale, and the game is fair. For $p = 0.6$ the added term is $0.2 > 0$, so X is a submartingale. For $p = 0.4$ it is $-0.2 < 0$, so X is a supermartingale.

The unconditional means follow. Each toss adds $2p - 1$ on average, so $\mathbb{E}[X_t] = (2p - 1)t$. That is 0 for the fair coin, $0.2t$ for $p = 0.6$ and $-0.2t$ for $p = 0.4$. Only the fair game leaves the mean where it started.

Whether a process is a martingale depends on *both* the filtration and the measure. That flexibility is the whole point. The stock in Example 2.11 is *not* a martingale under the physical measure, because $\mathbb{E}[S_{t+1} | \mathcal{F}_t] = 1.04 S_t > S_t$. Under the physical measure the stock has a positive expected excess return. In an equilibrium interpretation that is the risk premium investors require for bearing the stock's risk. It becomes a martingale when the probabilities are reweighted to $q = \frac{1}{2}$, since $\frac{1}{2}(1.2) + \frac{1}{2}(0.8) = 1$. Reweighting probabilities to remove drift is what the risk-neutral measure of Section 1 did in one period.

Proposition 2.14.

A martingale has constant expectation: $\mathbb{E}[X_t] = \mathbb{E}[X_0]$ for all t . If in addition $\mathcal{F}_0$ is trivial, then X_0 is a constant and $\mathbb{E}[X_t] = X_0$.

A fair game creates no expected gain at any horizon. The converse fails, and the failure matters. Constant expectation is a statement about unconditional averages only. It does not force the increments to be fair conditionally, which is what the martingale property demands. So constant expectation is necessary and not sufficient.

Proof.

□

Example 2.15. Stay with the die filtration of Example 2.4, with X the number rolled. Define $M_t = \mathbb{E}[X | \mathcal{F}_t]$ for $t = 0, 1, 2, 3$. Show that M is a martingale.

Write down the four dates. At $t = 0$ nothing is known, so $M_0 = \mathbb{E}[X] = 3.5$. At $t = 1$ and $t = 2$ Example 2.8 gives the values already computed. At $t = 3$ the roll is known, so $M_3 = X$:

$$M_0 = 3.5$$

$$M_1 = 3 \text{ on } \{1, 3, 5\}, \quad 4 \text{ on } \{2, 4, 6\}$$

$$M_2 = 2 \text{ on } \{1, 3\}, \quad 5 \text{ on } \{5\}, \quad 2 \text{ on } \{2\}, \quad 5 \text{ on } \{4, 6\}$$

$$M_3 = X$$

Now check the three steps. From date 0 to date 1, $\mathbb{E}[M_1] = \frac{1}{2}(3) + \frac{1}{2}(4) = 3.5 = M_0$. From date 1 to date 2, take the cell $\{1, 3, 5\}$: there $\mathbb{E}[M_2 | \{1, 3, 5\}] = \frac{2}{3}(2) + \frac{1}{3}(5) = 3 = M_1$, and $\{2, 4, 6\}$

works the same way. From date 2 to date 3, take $\{1, 3\}$: there $\mathbb{E}[M_3 \mid \{1, 3\}] = \frac{1+3}{2} = 2 = M_2$. On the cells $\{5\}$ and $\{2\}$ there is nothing left to average, since M_2 already equals X there.

Nothing in this depends on the die, and the general statement is the most important link between §2.2 and §2.3. Every integrable random variable generates a martingale, namely its own forecast process. For any X the process $M_t = \mathbb{E}[X \mid \mathcal{F}_t]$ satisfies

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] = \mathbb{E}[X \mid \mathcal{F}_t] = M_t,$$

where the middle step is the tower property, Proposition 2.9(c). Read it as a statement about learning. The forecast of a fixed quantity moves only as information arrives, and it never drifts in a predictable direction. Section 2.4 uses exactly this process, with X the payoff of a derivative and the measure reweighted to Q .

2.4 Multi-Period Markets and the FTAP

This subsection upgrades Section 1's fundamental theorem to markets that trade at every date. The statement changes in exactly one way. "Positive state prices exist" becomes "an equivalent measure exists under which discounted prices are martingales."

Fix the filtration of an event tree. Let S_t be the vector of risky asset prices, which is adapted. A riskless account paying the riskless rate r is worth $B_t = (1+r)^t$. A *trading strategy* θ_t is the portfolio chosen at date t , after $\mathcal{F}_t$ has been observed, and held into date $t+1$. So θ_t is $\mathcal{F}_t$ -measurable. You choose today's position with today's information and nothing later.

Prices are measured in units of the money-market account from here on. Write

$$\tilde{S}_t = \frac{S_t}{B_t}.$$

The reason is economic. Even in a market with no risk premium, a traded asset still grows at the riskless rate, so its raw price cannot be a fair game. What can be a fair game is its price expressed in units of the account that earns that rate. Discounting strips out the growth everyone gets for waiting, and leaves the part that compensates risk. The martingale condition below is imposed on $\tilde{S}_t$, never on S_t .

One remark on indexing, because other texts differ. Many of them write θ_t for the portfolio held from $t-1$ to t , which makes θ_t measurable with respect to $\mathcal{F}_{t-1}$. A process with that property is called *predictable*, and in that convention the slogan is that prices are adapted and strategies are predictable. We index by the date the portfolio is chosen, so the same restriction reads θ_t is $\mathcal{F}_t$ -measurable. The economics is identical.

A strategy is self-financing when every change in portfolio value comes from gains and losses on the assets already held. No outside cash is put in and none is taken out. Rebalancing is allowed, but it must be paid for by selling something in the portfolio.

Definition 2.16 (Self-financing strategy).

A strategy is self-financing if all rebalancing is done at market prices with no cash added or withdrawn: at every date t and in every state, the value of the portfolio carried into t equals the cost of the portfolio carried out of t ,

$$V_t = \theta_{t-1}^\top \bar{S}_t = \theta_t^\top \bar{S}_t \quad (t = 1, \dots, T-1),$$

where $\bar{S}_t$ stacks the risky prices and the riskless account. The two ends are not rebalancing dates. At the start $V_0 = \theta_0^\top \bar{S}_0$ is the cost of setting the position up. At maturity $V_T = \theta_{T-1}^\top \bar{S}_T$ is what the last position liquidates for.

An *arbitrage* is a self-financing strategy with $V_0 \leq 0$ and $V_T \geq 0$ in every state, with at least one strict inequality. That is, either $V_0 < 0$, or $V_T > 0$ in at least one state. This is the multi-period version of the free lunches of Definition 1.3. The two types there are the two ways the strict inequality can hold. Type I is $V_0 = 0$ with $V_T > 0$ somewhere: nothing paid today, something received later. Type II is $V_0 < 0$ with $V_T \geq 0$: money received today, nothing owed later.

Definition 2.17 (Equivalent martingale measure).

A measure Q is an equivalent martingale measure (EMM) if (i) Q is equivalent to the physical measure $\mathbb{P}$, written $Q \sim \mathbb{P}$, which means $Q(A) = 0$ if and only if $\mathbb{P}(A) = 0$; and (ii) discounted prices S_t/B_t are Q -martingales: $\mathbb{E}^Q[S_{t+1}/B_{t+1} \mid \mathcal{F}_t] = S_t/B_t$.

Condition (i) is where the word *equivalent* comes from. It says that $\mathbb{P}$ and Q agree about which events are impossible, and about nothing else. Every physical state here has $\pi_\omega > 0$, so on this finite space (i) says exactly that $Q(\{\omega\}) > 0$ for every ω . Q may reweight the states as much as it likes. It may not declare a physically possible path impossible.

Two things about Q are worth saying plainly, because both are easy to get wrong. Q is not a forecast of real returns; nobody believes the stock will grow at the riskless rate. Q is a pricing measure, and the only job it has is to make discounted traded prices martingales.

There is also a direct line back to Section 1. In one period, positive state prices can be normalized into risk-neutral probabilities by dividing each one by their sum. In many periods the same normalization is carried out conditionally, at every node of the tree. The resulting one-step probabilities are then multiplied along each path, and that product is Q .

Theorem 2.18 (FTAP, multi-period).

A finite multi-period market admits no arbitrage if and only if an equivalent martingale measure exists.

The theorem converts a statement about every possible self-financing trading strategy into the existence of a single pricing measure. That is what makes it usable: an infinite search is replaced by one object to construct.

The reading is otherwise the one from Section 1. No free lunch means the same thing as a consistent system of positive “prices of probability”. The only new feature is where that consistency has to hold. It must now hold conditionally at every node of the tree.

Proof.

□

The binomial model is where all of this becomes concrete, and it is also the model that turns into a numerical method. No arbitrage there determines the risk-neutral up probability, and backward induction produces the replicating strategy. We postpone that calculation to Section 5, where the tree is calibrated to a continuous-time model and priced on a computer.

This section built one chain, and it is worth reading it off in order. $\mathcal{F}_t$ describes what is known at date t . The forecast $\mathbb{E}[\cdot | \mathcal{F}_t]$ uses that information to predict. A martingale is a process whose forecast equals its current value. The fundamental theorem says that an arbitrage-free market carries an equivalent measure Q under which discounted asset prices are martingales. Sections 3 and 4 rebuild the same chain with Brownian motion in place of the tree.

Exercises

Exercise 2.1. On the two-period tree with $S_0 = 100$, $u = 1.2$, $d = 0.8$, decide which of the following are $\mathcal{F}_1$ -measurable, and explain: (a) S_1 ; (b) S_2 ; (c) $\max(S_0, S_1)$; (d) $\mathbf{1}\{S_2 > 90\}$.

Exercise 2.2. On the same tree with physical up probability $p = 0.7$ per step, compute $\mathbb{E}[S_2 \mid \mathcal{F}_1]$ on each cell and verify the tower property numerically.

Exercise 2.3. One-period market: a bond with $r = 0$ and a stock with $S_0 = 100$ moving to one of three values $S_1 \in \{120, 100, 80\}$. Characterize all equivalent martingale measures. Is the market arbitrage-free? Is it complete? Reconcile with Section 1's incomplete market example.